

Validation of the NASA ROBIN helicopter model using the OpenFOAM HiSA solver

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Abstract

This study presents a high-fidelity numerical analysis of the unsteady Rotor-Body Interaction of the NASA ROBIN (ROtor Body INteraction) helicopter model using the open-source OpenFOAM HiSA solver. To accurately simulate the complex aerodynamics of forward flight, a custom cyclicPitchMotion solver was developed and integrated with a multi-motion sliding mesh technique, allowing for the simultaneous simulation of primary rotor rotation and individual blade cyclic pitching. The methodology was validated against experimental data for a forward flight condition of $\mu = 0.151$, where the solver successfully captured the periodicity and phase of unsteady pressure fluctuations across the fuselage, particularly in the nose and tail regions. While quantitative discrepancies in pressure amplitude occurred at the pylon due to the omission of the rotor hub and turbulence modeling limitations, the results qualitatively reproduced the complex vortex structures and interactional flow physics. This research establishes a robust, scalable, and general-purpose open-source framework suitable for the aerodynamic design and analysis of rotary-wing aircraft and Advanced Air Mobility (AAM) vehicles.

keywords : OpenFOAM HiSA, NASA ROBIN, Cyclic Pitch Motion, Sliding Mesh

1. Introduction

With the recent development of helicopters and future Advanced Air Mobility (AAM), attempts to precisely analyze the Rotor-Body Interaction during aerodynamic design using Computational Fluid Dynamics (CFD) are being widely made. In particular, complex unsteady flow phenomena such as the cyclic pitch of the rotor occurring during forward flight cause pressure fluctuations on the fuselage surface, which have a significant impact on flight performance and stability; therefore, accurate simulation of these phenomena is essential. To quantitatively verify these interactions, NASA developed the ROBIN (ROtor Body INteraction) model as a standard test configuration, which is widely utilized as a benchmark for comparison and validation between wind tunnel tests and CFD analysis [1-3].

Various numerical analysis studies using the ROBIN model have been conducted thus far. Petermann et al. [2] applied a high-order WENO scheme to analyze the rotor wake structure and fuselage pressure distribution and examined the impact of the rotor wake on structural vibrations. Xu and Zhang [4] verified unsteady aerodynamic performance during forward flight through the Chimera overset grid method. Steijl and Barakos [5, 6] proposed sliding mesh and sliding plane algorithms to secure both computational efficiency and physical accuracy. Kowarsch et al. [7] identified the impact of the rotor hub wake on rear fuselage vibrations (tail shake). Furthermore, Kubrak and Snyder [8] verified unsteady pressure fluctuations using the commercial software STAR-CCM+. Biava and Vigevano [9], as well as Ma et al. [10], improved numerical precision through interference effect analysis and the implementation of trimming analysis methods, respectively. Heise et al. [11] presented an efficient analysis plan for the initial design stage by applying an actuator disc model.

These existing studies have mainly utilized expensive commercial software or specifically developed in-house codes. However, with the recent improvement in the performance of computational resources, there is an increasing demand to utilize open-source frameworks that offer excellent scalability and accessibility for analysis. In this paper, we aim to analyze the unsteady rotor-body interaction of the NASA ROBIN model using the open-source-based OpenFOAM HiSA (High Speed Aerodynamic)[12] solver and verify its suitability. In particular, to precisely simulate the cyclic pitch motion of the rotor blades, we directly implemented a **cyclicPitchMotion** solver. We established a high-fidelity analysis workflow by combining this with the multi-motion (**multiMotion**) sliding mesh technique and aim to examine its effectiveness.

2. Main Body

2.1 OpenFOAM CyclicPitchMotion Solver

In helicopter aerodynamics, rotor control is achieved through two main mechanisms: **collective** pitch and **cyclic** pitch. Through these adjustments, the aircraft can transition between vertical flight and directional flight.

Collective pitch refers to changing the pitch angle of all rotor blades simultaneously and equally. When a pilot adjusts the collective, the total lift generated across the entire rotor disk changes, allowing the helicopter to ascend or descend. **Cyclic pitch** is a method of individually changing the pitch angle of each blade according to the rotation position of the rotor, i.e., the **azimuth angle** (ψ). This change creates a lift imbalance in the rotor disk, causing the disk to tilt. This tilting changes the direction of the thrust vector, enabling horizontal movement (forward, backward, or lateral flight).

The pitch angle of each blade can be expressed according to the azimuth angle (ψ) as follows:

$$\theta(\psi) = \theta_0 - \theta_{1c} \cos(\psi) - \theta_{1s} \sin(\psi) \quad (1)$$

- θ_0 : Collective pitch
- θ_{1c} : Lateral cyclic pitch
- θ_{1s} : Longitudinal cyclic pitch

To simulate the cyclic pitch motion of rotor blades in OpenFOAM, the **cyclicPitchMotion** solver was implemented based on the **oscillatingRotatingMotion** solver provided by OpenFOAM. This solver expresses the pitch angle θ according to simulation time (t) as a harmonic oscillation function.

$$\theta(t) = \alpha_0 + A \sin(\omega t + \phi) \quad (2)$$

Table 1. **cyclicPitchMotion Parameter Comparison**

Feature	oscillatingRotatingMotion (Base)	cyclicPitchMotion (Custom)
Governing Equation	$\theta(t) = A \sin(\omega t)$	$\theta(t) = \theta_0 + A \sin(\omega t + \phi)$
Pitch Angle (θ)	Current pitch angle	Current pitch angle
Offset (α_0)	N/A	Collective Pitch
Amplitude (A)	Amplitude angle vector	Cyclic Amplitude angle
Velocity (ω)	Radial velocity	Radial velocity

Feature	oscillatingRotatingMotion (Base)	cyclicPitchMotion (Custom)
Time (t)	Simulation time	Simulation time
Phase (ϕ)	N/A	Phase angle

The parameter values of the **cyclicPitchMotion** solver in Equation (2) and the parameters of OpenFOAM's **oscillatingRotatingMotion** solver are as shown in **Table 1**. At this time, the parameter values can be calculated using the values from Equation (1) as shown in Equation (3).

$$\begin{aligned}\theta(t) &= \alpha_0 + A \sin(\omega t + \phi) \\ A &= \sqrt{\theta_{1c}^2 + \theta_{1s}^2} \\ \phi &= \text{atan2}(-\theta_{1s}, -\theta_{1c})\end{aligned}\tag{3}$$

To apply the solver to the simulation, define the **cyclicPitchMotion** function within the **constant/dynamicMeshDict** file.

For example, if a rotor rotating in the +z axis (0, 0, 1) direction around the origin (0, 0, 0) performs a pitch motion in the +x axis (1, 0, 0) direction such as $\theta(\psi) = 10^\circ - 3^\circ \cos(\psi) - 2^\circ \sin(\psi)$, the **cyclicPitchMotion** parameters are as follows:

- $A = \sqrt{(-3)^2 + (-2)^2} = 3.60555^\circ$
- $\phi = \text{atan2}(-2, -3) = 3.7296 \text{ radian}$

The **constant/dynamicMeshDict** reflecting this is as follows:

```
dynamicFvMesh    dynamicSolidBodyMotionFvMesh;

solidBodyMotionFunctions
{
    blade1
    {
        solidBodyMotionFunction cyclicPitchMotion;
        origin    (0 0 0);
        axis      (1 0 0);    // Pitch motion axis at time = 0
        omega     209.4395;   // 2000 rpm
        amplitude 3.60555;    // Result of sqrt(3^2 + 2^2)
        phase     3.7296;     // Result of atan2(-2, -3)
        alpha0    10;         // Collective pitch 10 degrees
    }
}
```

The blade rotates through a 360° azimuth angle and undergoes periodic oscillation within a range from a minimum of $6.39445^\circ = (10 - 3.60555)$ to a maximum of $13.60555^\circ = (10 + 3.60555)$. The rotation speed is 209.4395 rad/sec , which corresponds to approximately 2000 RPM .

2.2 Sliding mesh superposition using the multi-motion solver

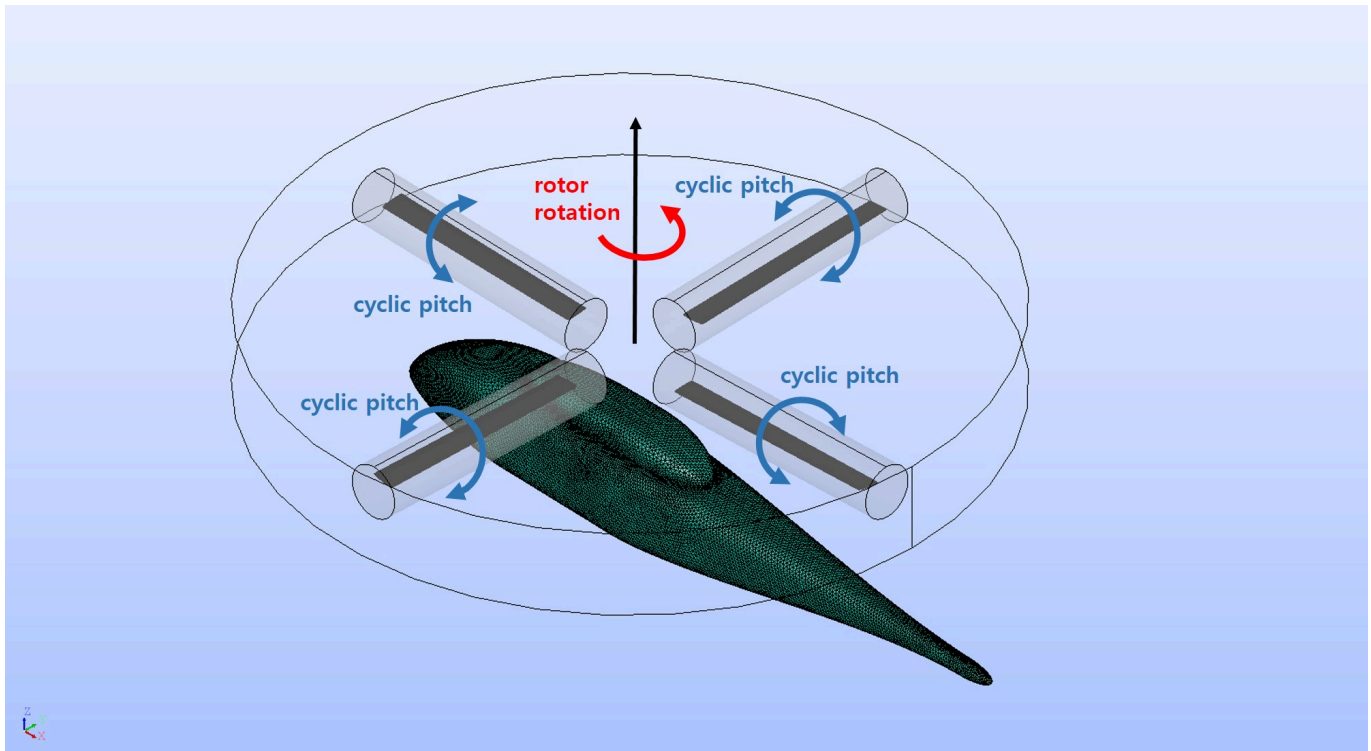


Fig. 1 Sliding mesh inside sliding mesh

For the cyclic pitch motion of a rotor composed of four blades, OpenFOAM's multi-motion (`multiMotion`) solver was applied so that as the sliding mesh containing the four blades rotates, each individual sliding mesh contained inside performs the cyclic pitch motion as shown in Fig. 1. At this time, the following two motions must be superimposed:

- **Primary Rotation** : The motion of the entire rotor disk rotating around the hub
- **Cyclic Pitch** : The motion of each individual blade pitching up and down as it rotates

For example, when the rotor rotates at 2000 RPM and the cyclic pitch motion of the blades is defined as $\theta(\psi) = 12^\circ - 2.9^\circ \cos(\psi) + 2.5^\circ \sin(\psi)$, the `cyclicPitchMotion` solver explained in the previous section is used for the cyclic pitch of individual blades, and the `rotatingMotion` solver is used for the rotation of the entire rotor to apply multi-motion (`multiMotion`).

At this time, `constant/dynamicMeshDict` is configured as follows. The phase difference of each individual blade has a difference of $\pi/2$ from the phase difference ϕ of the first blade. That is, the phase differences of the individual blades are ϕ , $\phi + \pi/2$, $\phi + \pi$, $\phi + 3/2\pi$. When the cyclic pitch axis of each individual blade has a coning angle of 1.7° , it is calculated relatively as the cyclic pitch axis vector at azimuth angles 0° , 90° , 180° , and 270° as the rotor rotates.

```
dynamicFvMesh dynamicMultiMotionSolverFvMesh

// Specification of rotation around centre
rotation
{
    solidBodyMotionFunction  rotatingMotion; // rotating motion
    rotatingMotionCoeffs
    {
        origin (0.696 0.051 0.322); // rotation center
        axis (0 0 1);                // rotation axis : +z
    }
}
```

```

    omega 209.4614;          // rotation speed : 2000 RPM
  }
}

dynamicMultiMotionSolverFvMeshCoeffs
{
  rotorzone // Primary rotation of total blades
  {
    solver solidBody;
    cellZone rotorzone;
    solidBodyCoeffs
    {
      $rotation;
    }
  }

  bladezone000 // 1st blade at azimuth angle 0
  {
    cellZone bladezone000;
    solver solidBody;
    solidBodyCoeffs
    {
      solidBodyMotionFunction multiMotion;
      multiMotionCoeffs
      {
        rotorzone // primary rotation
        {
          $rotation;
        }

        bladezone000 // 1st blade relative motion to the primary motion
        {
          solidBodyMotionFunction cyclicPitchMotion;
          cyclicPitchMotionCoeffs
          {
            origin (0.696 0.051 0.322); // pitch motion center
            axis (0.99955986 0 0.029666244); // relative pitch axis: +x
            omega 209.4614; // pitch motion speed
            amplitude 3.83; // amplitude
            phase 2.28; // 1st blade phase angle
            alpha0 12; // collective pitch
          }
        }
      }
    }
  }

  bladezone090
  {
    ...
    axis (0 0.99955986 0.029666244); // relative pitch axis: +y
    ...
    phase 3.850796; // 2nd blade phase angle
    ...
  }
}

```

```

}

bladezone180
{
    ...
    axis (-0.99955986 0 0.029666244); // relative pitch axis: -x
    ...
    phase 5.421592;                  // 3rd blade phase angle
    ...
}

bladezone270
{
    ...
    axis (0 -0.99955986 0.029666244); // relative pitch axis: -y
    ...
    phase 6.992389;                  // 4th blade phase angle
    ...
}
}

```

2.3 Analysis Model and Computational Mesh

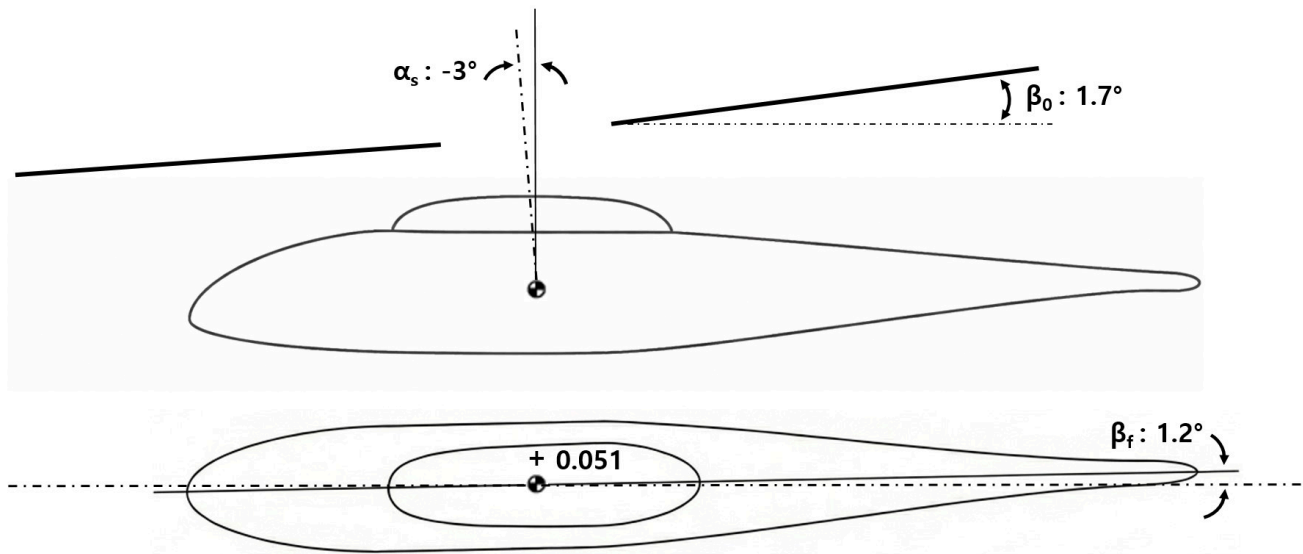


Fig. 2 Sketch of the ROBIN configuration

The model used in this study is the shape defined in the preceding research and is shown in Fig. 2. It consists of the main fuselage and a pylon corresponding to the engine fairing. The rotation direction of the rotor is counter-clockwise relative to the upper part, and the azimuth angle (ψ) 0° is located at the top rear of the fuselage. A fuselage yaw angle β_f 1.2° due to manufacturing errors mentioned in the experimental model and a rotor rotation center offset of 0.051 m in the +y direction from the fuselage center were applied. Additionally, an inclination angle α_s of -3° was applied to the rotor axis, and blade flapping was excluded while a coning angle β_0 1.7° was applied. The rotor blades use the **NACA 0012** airfoil and have an untapered shape with a linear twist angle of -8° . The chord length of the blade is 0.0663 m, the rotor radius is 0.86 m, and the root cutout is 24 %. The rotor hub was not considered.

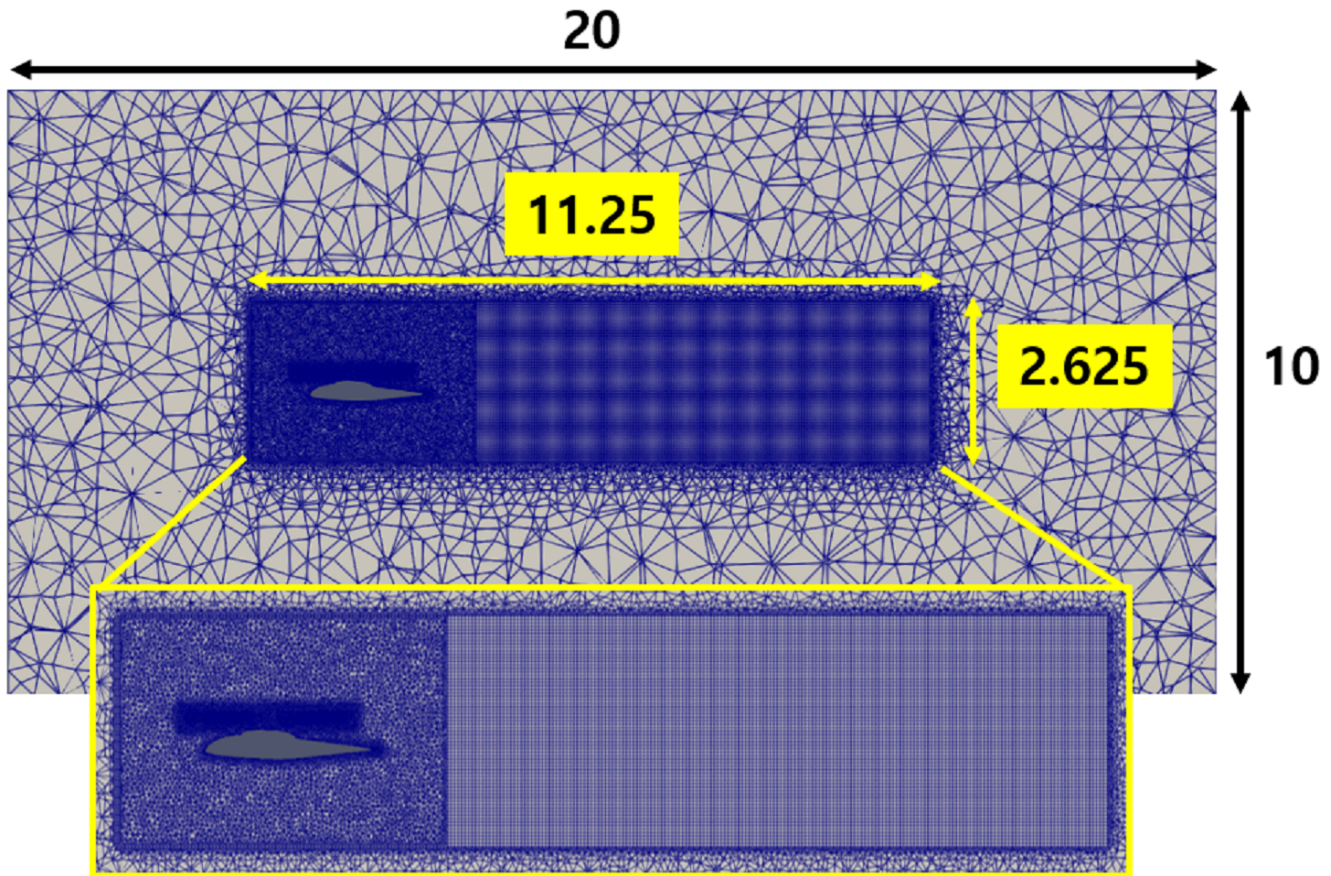


Fig. 3 Domain size and mesh

The size of the calculation domain and the mesh are shown in Fig. 3. Tetrahedral mesh was used for the calculation domain except for the fuselage, rotor, and wake regions, and a hybrid mesh[13] applying hexahedral mesh was used to reduce the number of meshes in the wake region.

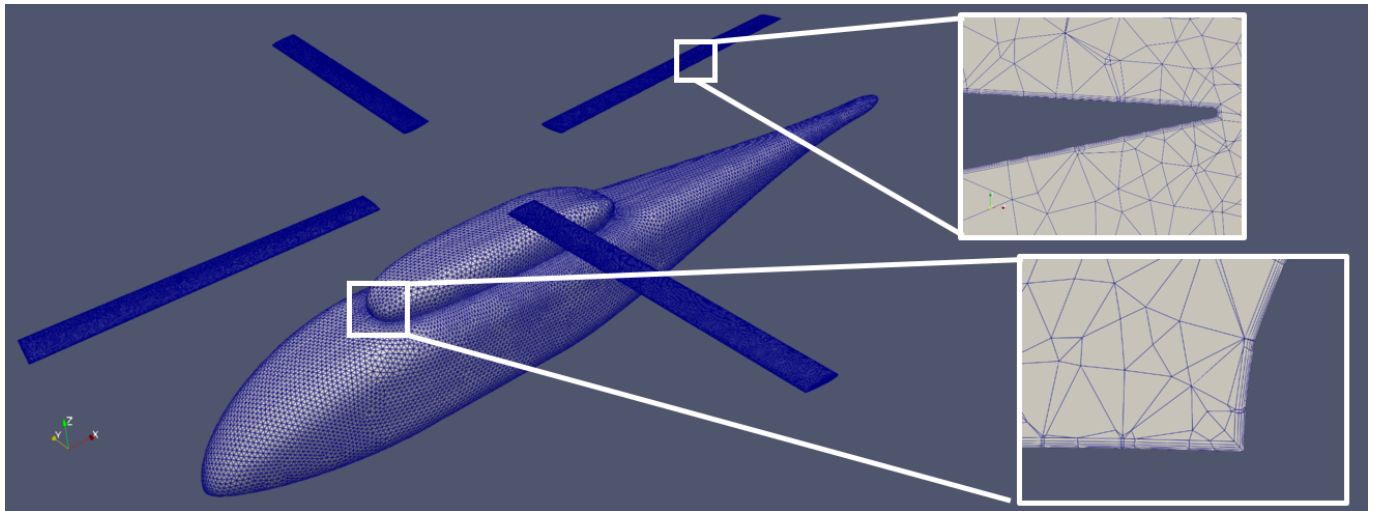


Fig. 4 Fuselage and rotor mesh and boundary layer mesh

The fuselage and rotor surface meshes are shown in Fig. 4. The size of the first mesh at the wall is 10^{-4} , and at this time, the y^+ value is about 10, ensuring no difficulty in applying the wall function boundary conditions of the $k - \omega$ SST turbulence model. A total of 5 cells were used in the boundary layer, giving a thickness of 10^{-3} . The total number of meshes is approximately 14 million. Model and mesh settings are summarized in Table 2.

Table 2. Model and mesh parameters

α_s (deg)	β_f (deg)	β_0 (deg)	rotor radius R (m)	root cutout (%)	twist (deg)	chord (m)	rotor center y (m)	first cell height (m)	y+	nCell
-3	1.2	1.7	0.86	24	-8	0.0663	0.051	10^{-4}	10	14 M

2.4 $\mu = 0.151$ Forward Flight Analysis

Among the experimental conditions of the preceding research, the analysis was performed for the $\mu = 0.151$ forward flight condition. The blade pitch angles are $\theta_0 = 12^\circ$, $\theta_{1c} = -2.9^\circ$, $\theta_{1s} = 2.5^\circ$, and the rotation speed is $\omega = 2000 \text{ RPM}$. The free-stream velocity is $V_\infty = 27.24 \text{ m/s}$, which corresponds to $M_\infty = 0.08$. At this time, the blade tip speed is $V_{tip} = 180.1368 \text{ m/s}$ and $M_{tip} = 0.525$, so compressibility effects must be considered. For unsteady flow analysis, the dual time-stepping technique of the HiSA solver was used, and a total of 10 rotations ($t = 0.2972$) were analyzed by applying 200 physical time intervals ($\Delta t = 0.0001486$) per rotor rotation. At this time, the rotation angle per physical time interval (Δt) is 1.8° . Analysis conditions are summarized in Table 3.

Table 3. simulation conditions

μ	V_∞ (m/s)	M_∞	V_{tip} (m/s)	M_{tip}	θ_0 (deg)	θ_{1c} (deg)	θ_{1s} (deg)	Ω (RPM)	Δt (sec)	rev.
0.151	27.24	0.08	180.1368	0.525	12	-2.9	2.5	2000	1.468^{-4}	10

2.4.1 Unsteady Pressure Distribution

In the reference, unsteady pressure was measured just to the starboard of the centerline on the top surface of the model and in the circumferential direction of the fuselage at the $x/L = 0.89$ position of the model; in the analysis, the measurement points in the experiment were adjusted to fit the analysis mesh. Due to numerical errors during mesh generation, there were cases where experimental measurement points were not located on the fuselage surface, so the nearest surface mesh cell was set as the measurement point. The exact coordinate values of the measurement points used in the analysis are shown in Table 4, and the locations are shown in Fig. 5. The pressure coefficient for the azimuth angle (ψ) at the 10th rotation was compared with the experimental values.

Table 4. Locations for unsteady pressure measurements in CFD

Top of body				Ring around body			
Orifice	x/L	y/L	z/L	Orifice	x/L	y/L	z/L
D5	0.053349	-0.006571	0.042693	D1	0.892457	-0.086491	-0.127651
D6	0.099028	-0.006717	0.073343	D3	0.901319	-0.113959	0.069097
D8	0.206353	-0.003622	0.117789	D4	0.90245	-0.102018	0.09512
D9	0.262313	-0.002507	0.132884	D19	0.902839	-0.073066	0.114189
D17	0.476904	0.001771	0.196746	D22	0.905009	0.010751	0.18934

Orifice	x/L	y/L	z/L
D18	0.610583	0.004535	0.206766
D22	0.905009	0.010751	0.18934
D26	1.008222	0.013092	0.133865
D14	1.184843	0.016971	0.084558
D15	1.371863	0.020958	0.061743
D16	1.558831	0.024948	0.03793

Orifice	x/L	y/L	z/L
D23	0.900927	0.080845	0.139617
D25	0.899618	0.0809	0.114652
D13	0.898216	0.107928	0.098762
D12	0.898757	0.120018	0.075751
D10	0.888696	0.098189	-0.125257

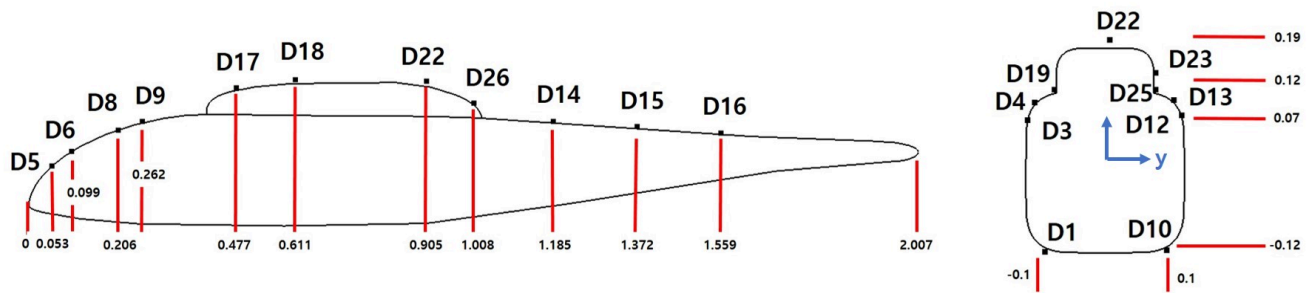


Fig. 5 top(left) and ring(right) locations for unsteady pressure measurements in CFD

The unsteady pressure coefficient C_p was non-dimensionalized by the dynamic pressure of the rotor tip as in the reference and multiplied by 100 for visualization, which is shown in Equation (4).

$$C_p = \frac{p - p_\infty}{\rho_\infty (\Omega R)^2} \times 100 \quad (4)$$

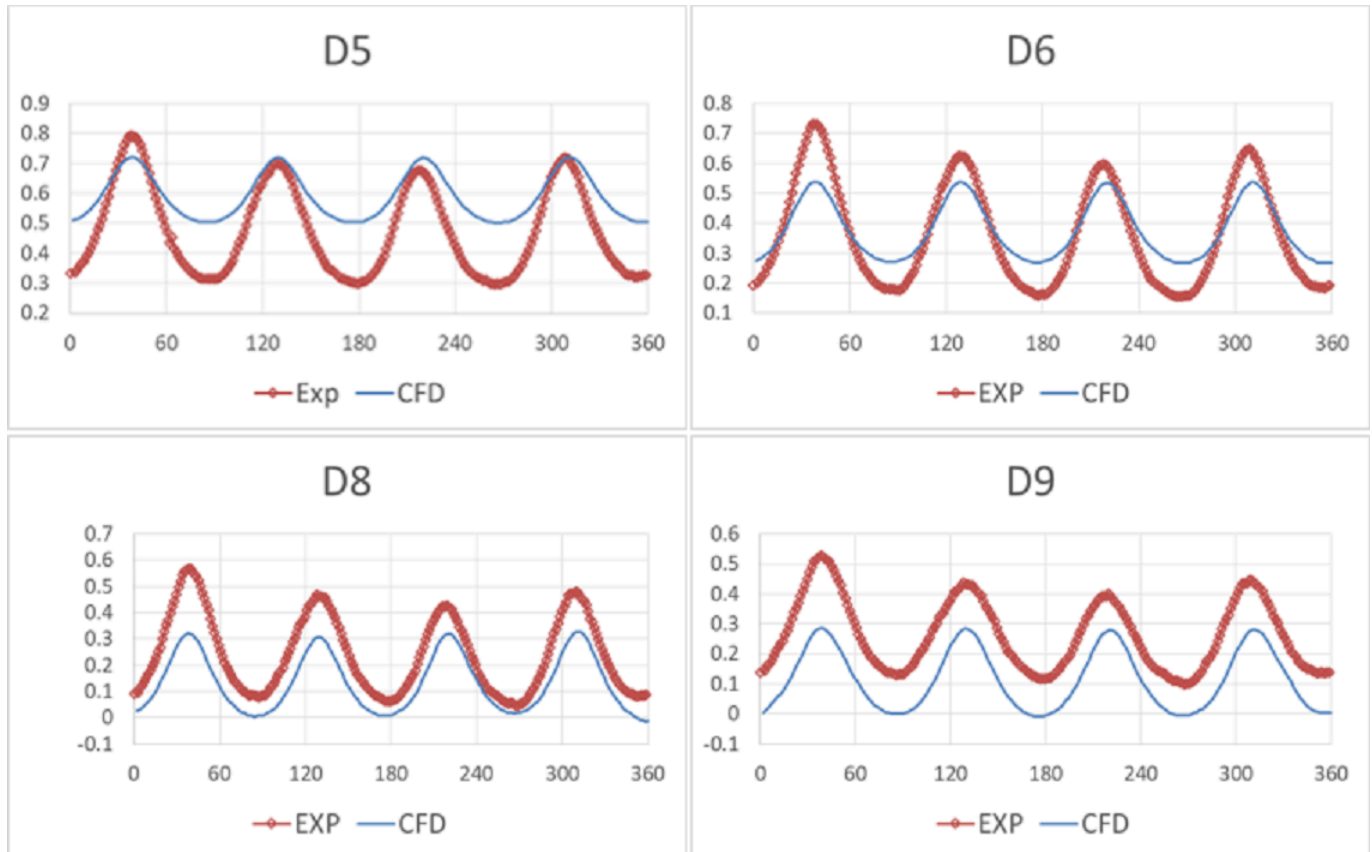


Fig. 6 Pressure coefficient over nose

Fig. 6 shows the pressure distribution at measurement points D5, D6, D8, and D9 in the nose part of the fuselage. The phase of the periodic pressure change at nose top measurement points D5~D9 matches the experiment well. At the frontmost D5, the maximum pressure value appeared similar to the experiment, but the amplitude was predicted to be low. At D6, the amplitude appeared smaller than the experiment, leading to the judgment that the vortex strength due to the blade passing was underestimated. At D8 and D9, the overall pressure level appears low, which is estimated to be due to the influence of mesh resolution in the area adjacent to the rotor's sliding mesh surface.

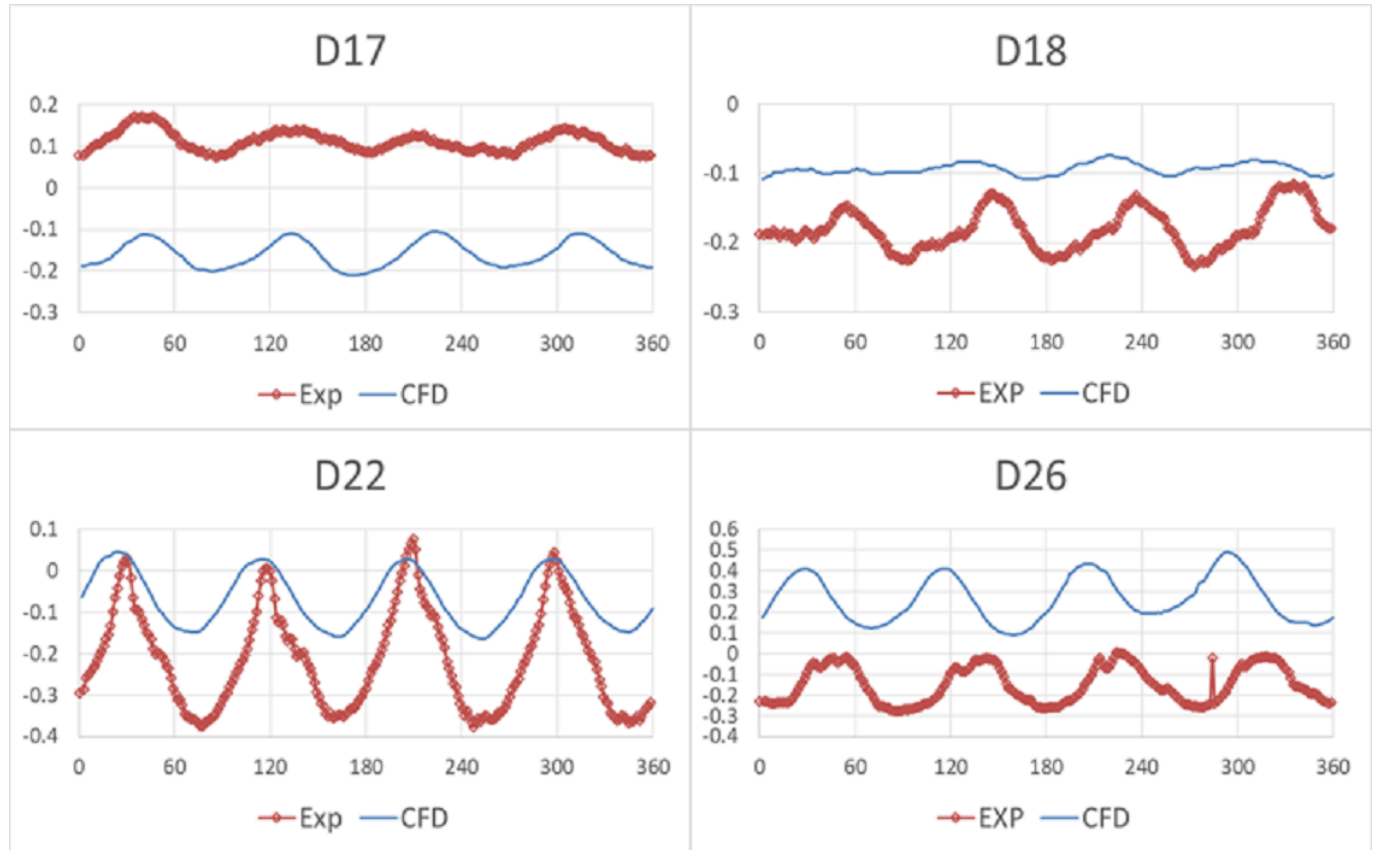


Fig. 7 Pressure coefficient over pylon

Fig. 7 shows the pressure change at the top of the pylon, where the pressure fluctuation at pylon top measurement points D17~D26 shows a complex flow structure where the root vortex and frontal vortex are superimposed. A strong vortex generated at the rotor root flows into the top of the pylon and combines with the vortex generated at the front part, causing unsteady pressure changes. The numerical analysis results captured the pressure fluctuation patterns according to blade rotation well but showed differences from the experiment in maximum pressure and amplitude. This is interpreted as a result of the root vortex excessively affecting the fuselage due to simplified modeling that does not consider the hub shape. Therefore, the top of the pylon is the most sensitive area in rotor-body interaction, suggesting that hub modeling and turbulence model improvement are necessary.

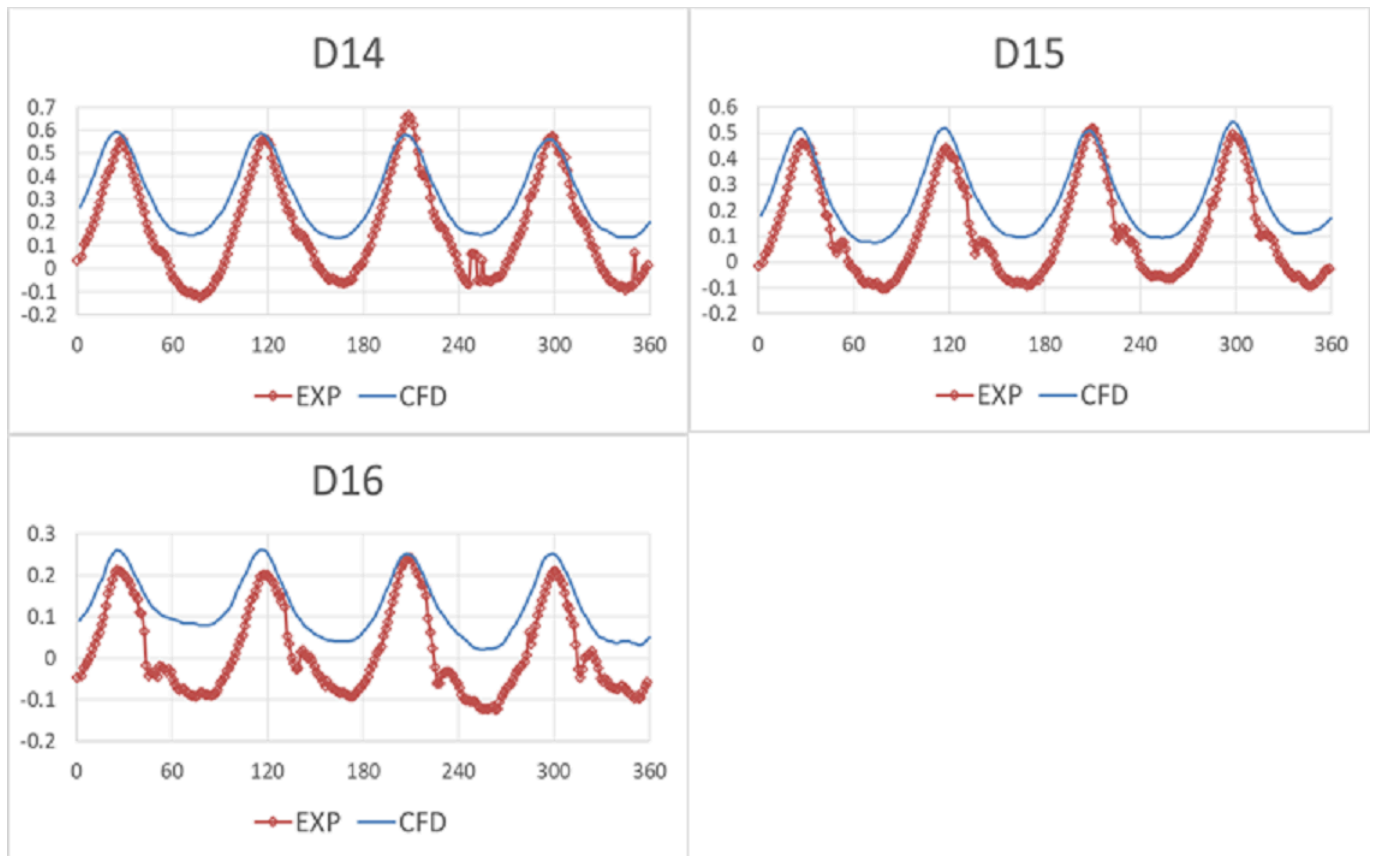


Fig. 8 Pressure coefficient over tail boom

The pressure distribution at rear measurement points D14~D16 in Fig. 8 shows the impact of the rotor wake on the rear of the fuselage. The vortex generated when the rotor blade passes the tail part causes periodic pressure changes on the rear surface; the pressure amplitude decreases, but the maximum pressure appears similar to the experiment. The CFD results showed small amplitudes, but the maximum values match the experiment well, showing a similar trend to previous research.

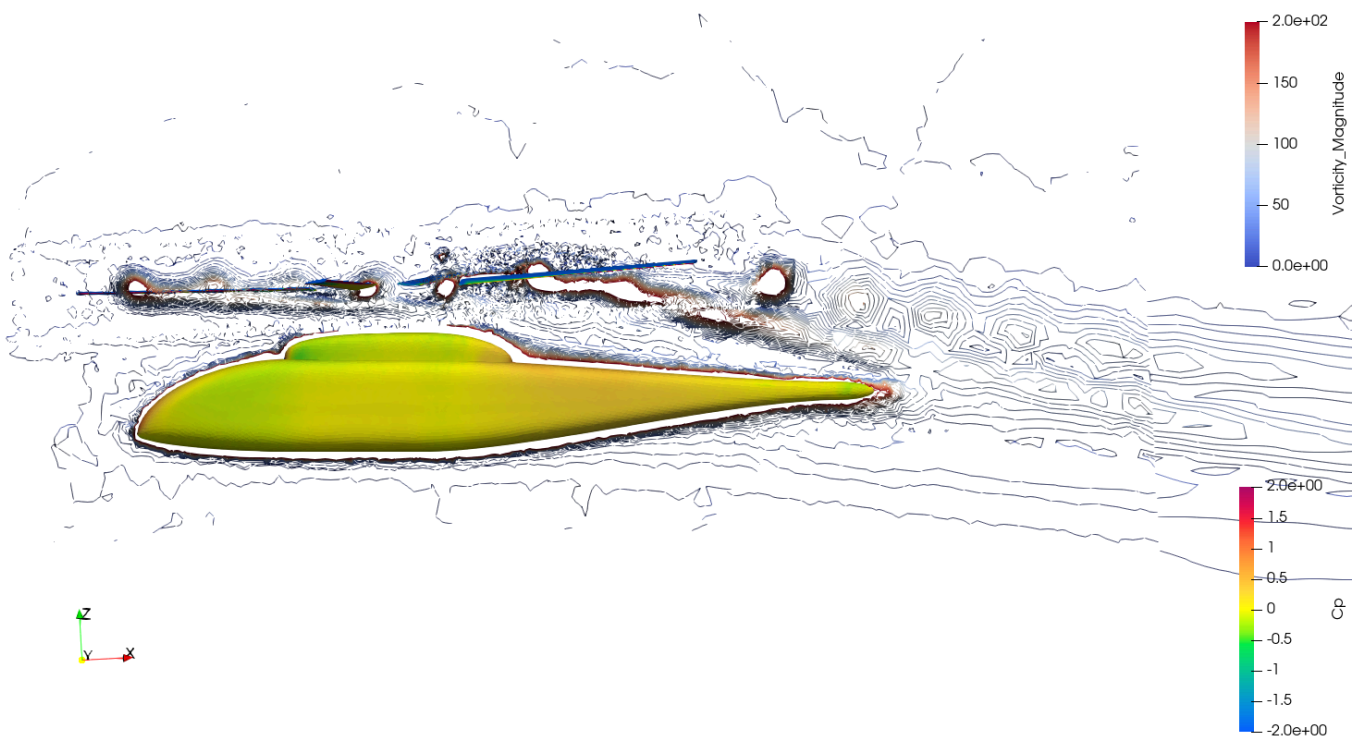


Fig. 9 Vorticity field at centerline ($y = 0$)

The vorticity distribution at the fuselage centerline ($y = 0$) is shown in Fig. 9, displaying a typical structure of rotor-body interaction. The vortex generated at the front part and the root tip vortex overlap at the top and bottom in the rear, forming a complex vortex structure. It can be seen that as the root tip vortex flows into the top of the pylon and combines with the front part vortex, it causes complex pressure changes at the top of the pylon, and it can be inferred that this will be greatly affected by the presence of a hub. It is judged that additional research on the impact of the hub is needed in the future.

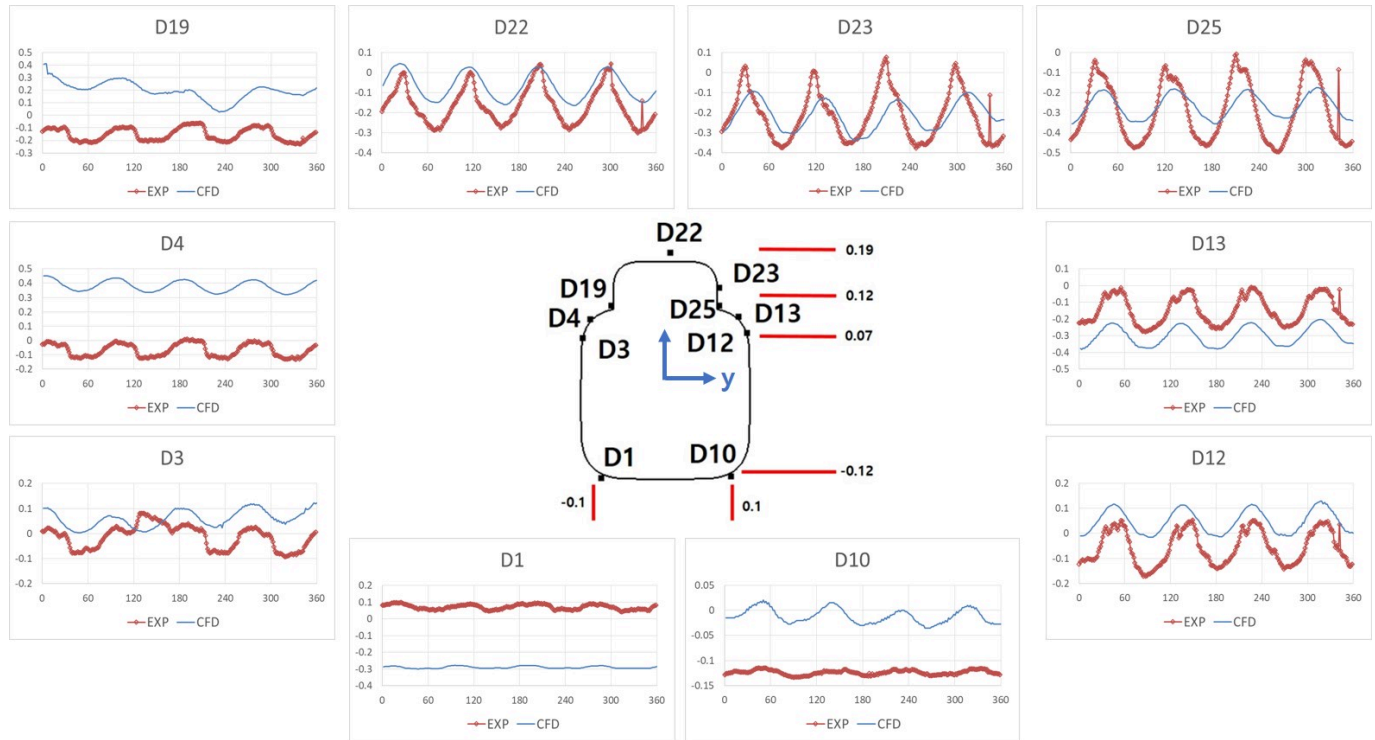


Fig. 10 Pressure coefficient around rin ($x/L = 0.89$)

Fig. 10 is the pressure distribution in the circumferential direction of the fuselage at the $x/L = 0.89$ position, showing a result combined with the rotor wake and fuselage shape effects. At the bottom and port side, the root tip vortex and separation flow combined to cause large pressure fluctuations, leading to differences with experimental results. On the other hand, at the top and starboard side, the pressure amplitude was predicted to be somewhat low, but the overall distribution showed a relatively similar trend compared to the experiment. These results are interpreted as the blunt fuselage shape causing separation flow easily, which interacts with the root vortex to cause inconsistency in pressure distribution. Therefore, it can be confirmed that the fuselage shape and vortex interaction have a decisive impact on the pressure distribution, and research on the impact of turbulence models and mesh resolution is necessary.

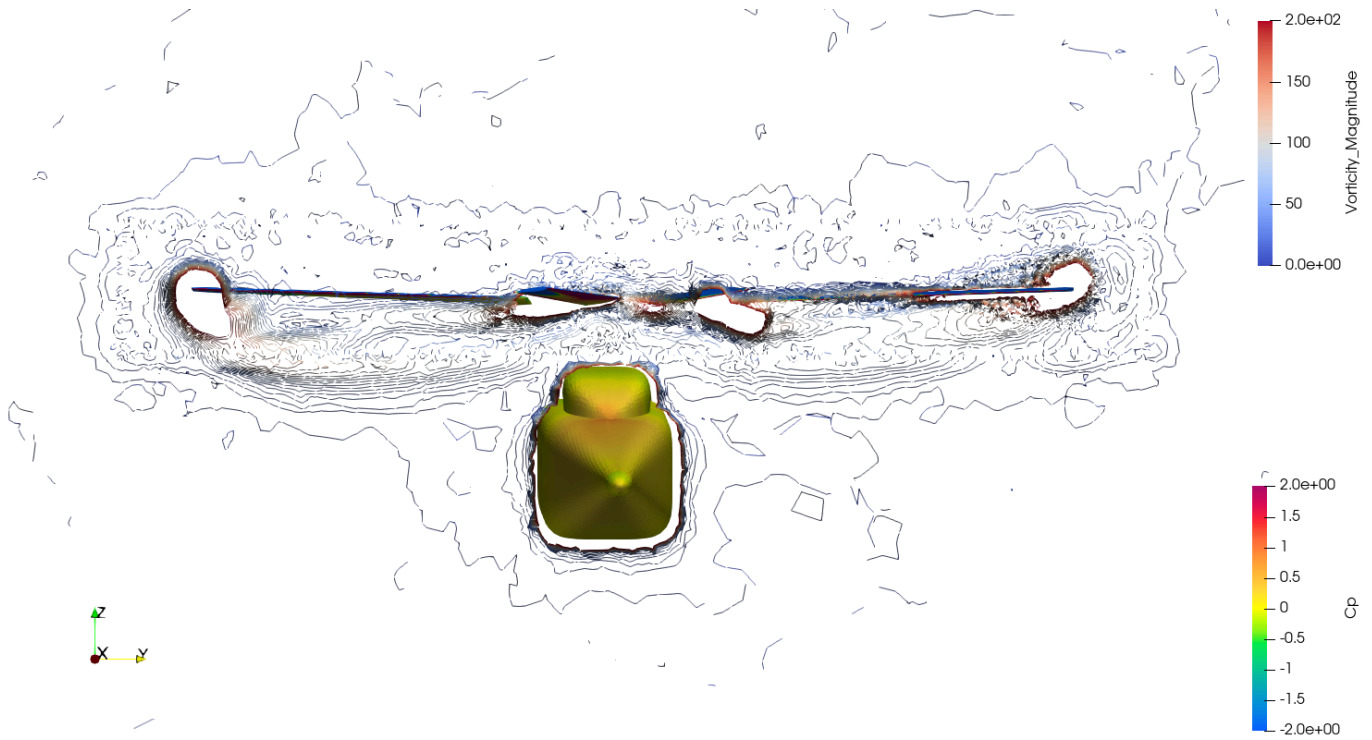


Fig. 11 Vorticity field at $x/L = 0.89$

Fig. 11 shows the vortex distribution at the $x/L = 0.89$ position. It can be confirmed that during forward flight, the difference in vortex strength between the retreating side and the advancing side is combined with the flow separation around the fuselage to form an asymmetric vortex structure. In particular, a strong vortex generated at the blade tip is clearly observed in the wake region, which clearly shows the major characteristics of rotor-body interaction and allows quantitative confirmation of the impact of the rotor tip vortex on the fuselage wake structure.

2.4.2 Steady Pressure Distribution

Fig. 12 shows the average pressure coefficient (C_p) distribution measured along the top centerline of the fuselage compared with experimental data and previous research results.

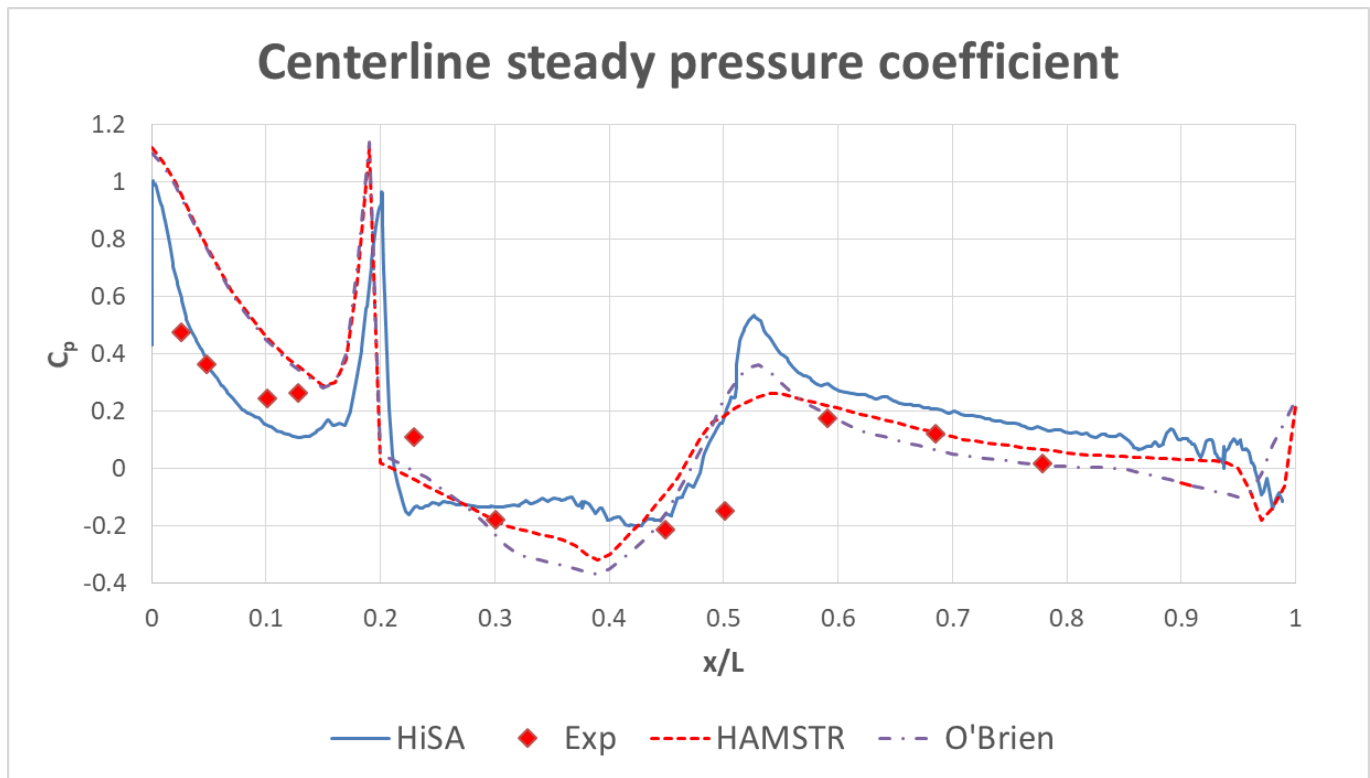


Fig. 12 Fuselage centerline steady pressure coefficient

The overall pressure distribution trend appears quite similar to the experimental values, and in particular, it well captures the sudden pressure drop occurring at the pylon forehead at the $x/L = 0.2$ point. However, at the top of the pylon and the tail area, a trend where the pressure is predicted to be slightly higher than experimental values and other numerical analysis results is observed; it is judged that since the hub was omitted, local pressure recovery appeared stronger due to the influence of the root vortex. The over-prediction of pressure in the tail area is related to the prediction precision of the flow separation point occurring at the rear of the fuselage, which is judged to be significantly influenced by the turbulence model. Despite these quantitative differences, the fact that the pressure gradient across the entire fuselage and the locations of peak points are in good agreement with experimental values confirms that the results of this analysis reasonably simulate the basic aerodynamic interference phenomena between the rotor and the fuselage.

3. Conclusion

In this study, based on the OpenFOAM HiSA solver, we implemented a **cyclicPitchMotion** solver that can simulate the cyclic pitch motion of rotor blades, and by combining this with the multi-motion (**multiMotion**) function and sliding mesh technique, we analyzed the unsteady rotor-body interaction of the NASA ROBIN model. By applying the implemented analysis technique to the forward flight condition ($\mu = 0.151$), we verified the suitability of the analysis method by comparing the unsteady pressure distribution and average pressure distribution on the fuselage surface with experimental and previous research results.

By directly simulating the motion of rotor blades including cyclic pitch motion, we confirmed that the phase of pressure change at major measurement points on the fuselage and the azimuth angle where maximum and minimum pressures occur generally match well with experimental results and existing research. In particular, in the nose and rear areas, the periodicity of pressure distribution and peak positions were relatively accurately predicted, confirming that this analysis technique reasonably simulates basic unsteady interference phenomena between the rotor wake and the fuselage. Additionally, through the vorticity distribution, we

confirmed that the process where rotor tip vortices and root vortices interact with the flow around the fuselage to form complex vortex structures is qualitatively well reproduced.

Differences from experimental values appeared in pressure amplitude and average pressure level in some areas; such quantitative errors are phenomena commonly reported in existing ROBIN model analysis studies and are judged to be mainly due to the omission of the hub shape, mesh resolution, and limitations of the turbulence model. In particular, the top of the pylon and the rear area of the fuselage were identified as areas sensitive to the interaction between root vortices and separation flow, requiring additional modeling improvements.

We confirmed that the `cyclicPitchMotion` solver and multi-motion-based sliding mesh technique implemented in this study can simultaneously and stably simulate the primary rotation of the rotor and the cyclic pitch motion of the blades in the OpenFOAM environment, and it is significant in providing a general-purpose analysis framework applicable to complex rotary-wing flow analysis.

In future research, it is necessary to improve quantitative prediction accuracy through more detailed rotor modeling including the hub shape, convergence verification according to mesh resolution, and the application of high-precision turbulence models such as DES/LES. Furthermore, if extended to rotor analysis combined with trim analysis, it is expected to develop into a high-fidelity analysis tool usable in the design stage of actual rotary-wing aircraft and AAM vehicles.

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